

Question	Answer	Marks												
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Question	Answer	Marks
6(b)	Four from (max 3 marks for benefits only, without an explanation): • More read/write cycles (over its lifetime) // greater longevity ... • ... likely to be a lot of read/write functions each day • Read/write speed is sufficient ... • ... even though it is slower than solid-state • Cheaper per unit of data stored ... • ... better value for the company to purchase • ... so the law company can afford to buy a server with greater storage capacity • No requirement for portability ... • ... as a server, it does not need to be moved • Trusted technology ... • ... it has been traditionally used for many years	4
6(c)	• DVD • CD • Blu-ray	3

Question	Answer	Marks
1(a)	Three from e.g.: – Keyboard – Mouse – Microphone – 2D scanner – 3D scanner – Touchscreen – Webcam // digital camera – Joystick – Trackpad – Sensor – Interactive whiteboard	3
1(b)	Three from e.g.: – Monitor // touchscreen – Inkjet printer – Laser printer – 3D printer – Speaker – Headphones – LED Projector – DLP – 2D cutter – 3D cutter – Actuator	3

Question	Answer	Marks
2(a)	Two from: – It is non-volatile – Can be easily disconnected from the computer – It is not directly accessed by the CPU – Suitable example e.g. CD, DVD, USB flash memory	2
2(b)(i)	One from: – ROM – RAM	1
2(b)(ii)	Two from: – HDD – SSD – Flash memory	2

Question	Answer	Marks
3	1 mark for each correct term, in the correct place: – Data/instructions – Instructions/data (must be the alternative to MP1) – Fetched – RAM – Decoded – Executed	6

Question	Answer	Marks
4(a)	Two from e.g.: – <u>DVD</u> uses red laser/light whereas <u>blu-ray</u> uses blue/violet laser/light – <u>DVD</u> has a smaller (storage) capacity // <u>Blu-ray</u> has a larger (storage) capacity – <u>DVD</u> has two layers (of polycarbonate) whereas <u>Blu-ray</u> disks have a single layer (of polycarbonate) – <u>DVD</u> has a slower transfer rate (of approximately 10 mbps) // <u>Blu-ray</u> has a faster transfer rate (of approximately 36 mbps)	2

4(b)(ii)	Four from: – Rollers are used to move the paper through the printer – Nozzles spray/drop/jet ink onto the paper – Ink jets/print head/nozzles are moved across the paper (to distribute the ink) – Different colour inks are mixed to create required colours – Technology could be piezoelectric – Technology could be thermal bubble – Ink is heated – ... and expands/evaporates into a bubble – Bubble is pushed through the nozzle on to the paper – ... then the bubble collapses – Electrical current is applied to a crystal – ... which makes it vibrate – ... which forces a droplet of ink through the nozzle	4
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Question	Answer	Marks
7(a)	Six from: – Motion sensor is used – Sensor sends data/signal to microprocessor – Data/Signal is converted from analogue data to digital data (using ADC) – Value to compared to stored value(s) // – If value is outside range/matches ... – ... microprocessor sends signal to switch lights on – ... actuator used to switch light on/off – ... timer is set for 2 minutes – Every time movement is detected the timer is reset – When timer reaches 0/120/times out microprocessor sends signal to switch lights off – Process is continuous	6
7(b)	Three from: – Read only memory – Non-volatile memory // Contents of memory are retained when power is turned off/permanent storage – Primary storage // directly accessed by the CPU – Holds firmware/boot-up instructions/start-up instructions/BIOS – Cannot be written to	3

Question	Answer	Marks
10	Four from: – Electrical field/charge is spread across the screen – Sensors are located around the screen // sensors are used to read the electric field – When finger touches screen, the charge/ is transferred to the user – ... as it is affected by the conductivity of another object – Coordinates of touch determined/calculated/measured	4

Question	Answer	Marks
6(a)	Four from: – Screen has two / multiple layers – Visitor presses on top layer – Top layer connects to bottom layer – ... creating a circuit – Calculation is carried out on where layers are connected	4
6(b)	Two from: – Speaker – Headphones – Printer	2

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Question	Answer	Marks
6(c)	Four from (max. 2 marks per type): Primary – Memory that is directly accessed by the CPU – An example is RAM / ROM – RAM stores programs and data that are currently in use and ROM stores boot-up instructions – RAM is volatile and ROM is non volatile Secondary – Storage that is not directly accessed by the CPU – An example is HDD / SSD – Stores data / files that can be accessed at a later stage – Non volatile	4

Question	Answer	Marks
1(a)	Printer Statement <pre> graph LR IP[Inkjet printer] --- C[Can print in colour] IP --- L[Creates output line by line using a print head] LP[Laser printer] --- C LP --- D[Uses a charged drum to create the printed item] LP --- T[Uses powdered toner] LP --- L </pre> One mark for correct lines from inkjet One mark for correct lines from laser	2
1(b)	α Laser	1
1(c)	Two from: ∞ Design is created on the computer / software / CAD ∞ Material is loaded to cutter ∞ Different types of material can be used ∞ Uses lasers to cut material ... ∞ ... that use infra-red ∞ ... that produces extreme heat ∞ ... that is focussed using a special lens ∞ Can work on both the x,y and z axis	2

Question	Answer	Marks
4(a)	One from: ∞ Touch screen ∞ Keyboard ∞ Microphone ∞ Mouse	1
4(b)	One from: ∞ Headphones ∞ Speakers ∞ Printer ∞ Light / LED	1
4(c)	One from: ∞ HDD ∞ SSD ∞ USB drive	1
4(d)	Four from: ∞ QR code is scanned using a <u>camera</u> on a mobile device ... ∞ ... and read / decoded using an application / software ∞ Illuminator shone on code ∞ Squares reflect light differently ∞ Corners of code are used for orientation ∞ Opens document with information // Directs to website with information ∞ QR code can be saved for future reference	4

7(d)	Two from: ∞ Can work continuously ∞ Avoids human error ∞ It could be a dangerous environment and will avoid human risk ∞ Detect errors instantly ∞ Maintain consistent and correct conditions	2
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Question	Answer	Marks
8	Six from: ∞ PC holds address of the instruction ∞ The address held in PC is sent to MAR ... ∞ ... using address bus ∞ MAR goes to location in memory where instruction is stored ∞ Instruction sent to MDR ... ∞ ... using data bus ∞ Instruction sent to CIR ∞ Control unit sends signals to manage the process ... ∞ ... using the control bus	6

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Question	Answer	Marks
2	Four from: ∞ Arithmetic and logic unit (ALU) ∞ Memory address register (MAR) ∞ Memory data register (MDR) // Memory buffer register (MBR) ∞ Accumulator (ACC) ∞ Immediate Access Store (IAS) ∞ Main memory // RAM ∞ Program counter (PC) ∞ Current instruction register (CIR) ∞ Address bus ∞ Data bus ∞ Control bus ∞ Input device ∞ Output device ∞ Secondary storage device	4

Question	Answer	Marks																					
6	One mark for each correct tick <table border="1" style="width: 100%; border-collapse: collapse; text-align: center;"> <thead> <tr> <th style="width: 60%;">Statement</th><th style="width: 20%;">Resistive (✓)</th><th style="width: 20%;">Capacitive (✓)</th></tr> </thead> <tbody> <tr> <td>This touch screen has multi-touch capabilities</td><td></td><td>✓</td></tr> <tr> <td>This touch screen cannot be used whilst wearing gloves</td><td></td><td>✓</td></tr> <tr> <td>This touch screen is made up of two layers with a small space in between</td><td>✓</td><td></td></tr> <tr> <td>This touch screen uses the electrical properties of the human body</td><td></td><td>✓</td></tr> <tr> <td>This touch screen is normally cheaper to manufacture</td><td>✓</td><td></td></tr> <tr> <td>This touch screen has a quicker response time</td><td></td><td>✓</td></tr> </tbody> </table>	Statement	Resistive (✓)	Capacitive (✓)	This touch screen has multi-touch capabilities		✓	This touch screen cannot be used whilst wearing gloves		✓	This touch screen is made up of two layers with a small space in between	✓		This touch screen uses the electrical properties of the human body		✓	This touch screen is normally cheaper to manufacture	✓		This touch screen has a quicker response time		✓	6
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Question	Answer	Marks
7(a)	Four from: ∞ Membrane / matrix / circuit board present at base of keys ∞ A key is pressed that presses a switch ∞ When a key is pressed it completes a circuit // changes the current in a circuit ∞ The location of the keypress is calculated ∞ An index of characters is searched to find the corresponding keypress ∞ Each character has an ASCII / Unicode value ∞ The ASCII / Unicode value has a binary value ∞ Keypress generates an interrupt ∞ Each character / keypress is added to a buffer to wait to be processed ∞ The binary can then be processed by the CPU to action the key press	4

Question	Answer	Marks
11(a)	RAM ∞ To store the data / instructions / parts of OS that are currently in use ROM ∞ To store the firmware / bootup instructions / BIOS SSD ∞ To store files / software // by example	3
11(b)	Two from: ∞ It is more durable // it has no moving parts ∞ It has a faster read / write / access speed ∞ It is more compact / light weight / smaller / portable ∞ It uses less energy // battery will last longer ∞ It is quieter ∞ Not affected by magnetic forces ∞ It runs at a cooler temperature ∞ Less latency // takes less time to warm up	2

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Question	Answer	Marks
1(a)(i)	Two from: ∞ 2D scanner ∞ Touchscreen ∞ Keypad/keyboard ∞ Card reader ∞ Mouse ∞ Digital camera	2
1(a)(ii)	Two from: ∞ HDD ∞ SSD ∞ USB flash memory drive ∞ SD card ∞ Any optical	2
1(a)(iii)	Two from: ∞ Monitor/Touch screen ∞ Speaker ∞ Printer ∞ LED // Light	2

1(d)	Five from: ∞ The display is made up of pixels ... ∞ ... that are arranged together as a matrix ∞ Each pixel has three filters, red, blue and green ∞ Shades of colour are achieved by mixing red, blue and green ∞ The screen is backlit ∞ Light is shone through the liquid crystals ∞ The liquid crystals can be made to turn solid or transparent/on or off ... ∞ ... by changing the shape of the crystal	5
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4(c)	Six from: ∞ Sensor sends a signal/reading/data to the microprocessor ∞ Signal/reading/data is analogue and is converted to digital using ADC ∞ Reading/data is stored in the system ∞ Microprocessor compares data/reading to the pre-set value of 7 ∞ If value is greater than 7 ... ∞ ... a signal/data is sent by the microprocessor to display a warning message on a monitor ∞ The process is continuous	6
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Question	Answer	Marks
7(a)(i)	Three from: ∞ RAM ∞ Primary memory ∞ Volatile memory ∞ Holds currently in use data/instructions ∞ Directly accessed by the CPU	3
7(a)(ii)	Two from: ∞ Arithmetic and logic unit (ALU) ∞ Memory address register (MAR) ∞ Memory data register (MDR) // Memory buffer register (MBR) ∞ Accumulator (ACC) ∞ Immediate Access Store (IAS) ∞ Control Unit (CU) ∞ Program counter (PC) ∞ Current instruction register (CIR) ∞ Address bus ∞ Data bus ∞ Control bus ∞ Input device ∞ Output device ∞ Secondary storage device	2

Question	Answer	Mark
1(a)	Component Control Bus Program Counter (PC) Memory Data Register (MDR) Accumulator (ACC) Description Increments to point to the address of the next instruction to be fetched Holds the result of a calculation. It is located within the Arithmetic Logic Unit (ALU) Carries signals to synchronise the fetch-execute cycle Temporary storage between the Central Processing (CPU) and primary memory 1 mark for 1 correct line 2 marks for 2 correct lines 3 marks for 3/4 correct lines	3
1(b)	Any two from: - Address bus - Data bus	2

Question	Answer	Mark
3(a)(i)	Any one from: - They are both non-volatile - They are both secondary storage // Both not directly accessed by the CPU - They both have a high capacity of storage - Both have read and write abilities	1
3(a)(ii)	Any three from: - HDD has moving parts but SSD does not - HDD uses magnetic storage whereas SSD uses flash memory - HDD is slower to access data than SSD // HDD has greater latency than SSD - HDD will create noise/heat, whereas SSD runs quieter/cooler - HDD has higher power consumption than SSD - HDD have greater longevity/more read-write cycles whereas SSD has lower longevity/limited number of read-write cycles - HDD larger in physical size/heavier than SSD - HDD is normally cheaper for the same capacity of storage as SSD - HDD is available in a larger storage capacity than SSD	3
3(b)	Any one from: - USB flash memory drive External HDD/SSD - SD Card - CD / DVD / Blu-ray	1
3(c)(i)	Any two from: - Keyboard - Mouse - Microphone - Touchscreen	2
3(c)(ii)	Any two from: - Monitor / Screen - Speakers - Headphones - Printer	2

Question	Answer	Mark
7	Any seven from: • Uses light sensor and Infrared / Motion / Pressure sensor • Sensors send data to the microprocessor • Data is converted from analogue to digital (using ADC) • Microprocessor compares both values to stored values • If motion value is out of range/in range, light value is checked // If light value is ≤ 10, motion value is checked • If light value is ≤ 10 lights are turned on // If motion value is out of range/in range lights are turned on ... • ... by sending a signal to actuator • Lights remain on for set period (and then turn off) // If motion is in range/out of range or light is > 10 then signal sent to turn lights off • Process repeats / is continuous	7

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Question	Answer	Marks
1(a)	Any three from: – MAR – MDR // MBR – PC // IAR // NIR // SCR – ACC – CIR // IR – IAS	3
1(b)(i)	– Fetch – Execute	2
1(b)(ii)	– Control unit	1

Question	Answer	Marks																					
6	<table border="1" style="width: 100%; border-collapse: collapse;"> <thead> <tr> <th style="width: 60%;">Statement</th><th style="width: 20%;">Capacitive (✓)</th><th style="width: 20%;">Resistive (✓)</th></tr> </thead> <tbody> <tr> <td>Needs pressure to be applied to create a circuit</td><td></td><td style="text-align: center;">✓</td></tr> <tr> <td>May not register a touch if the user is wearing gloves</td><td style="text-align: center;">✓</td><td></td></tr> <tr> <td>More commonly used in smartphones</td><td style="text-align: center;">✓</td><td></td></tr> <tr> <td>More responsive to a touch</td><td style="text-align: center;">✓</td><td></td></tr> <tr> <td>Needs an electrical field to be changed to register a touch</td><td style="text-align: center;">✓</td><td></td></tr> <tr> <td>Cheaper to manufacture</td><td></td><td style="text-align: center;">✓</td></tr> </tbody> </table> One mark per correct tick	Statement	Capacitive (✓)	Resistive (✓)	Needs pressure to be applied to create a circuit		✓	May not register a touch if the user is wearing gloves	✓		More commonly used in smartphones	✓		More responsive to a touch	✓		Needs an electrical field to be changed to register a touch	✓		Cheaper to manufacture		✓	6
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8(b)	Two marks for benefits, one mark for drawback Benefits: – Faster speed of printing – Can print duplex / on both sides – Many letters can be printed from one toner cartridge – Can print in high volumes Drawback – Toner cartridge more expensive to buy – More time to warm-up – Larger footprint	3
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Question	Answer				Marks
9(a)		Statement	HDD (✓)	SSD (✓)	6
		It has a limited number of read/write cycles		✓	
		It uses magnetic properties to store data	✓		
		It has moving parts	✓		
		It is non-volatile storage	✓	✓	
		It can be used as an external storage device to back-up data	✓	✓	
		It uses flash memory to store data		✓	
		9(b)	Any two from: – CD (drive) – DVD (drive) – Blu-ray (drive)		

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Question	Answer				Marks																												
2(a)	<table><tr><th>Hardware device</th><th>Input (✓)</th><th>Output (✓)</th><th>Storage (✓)</th></tr><tr><td>Solid state drive (SSD)</td><td></td><td></td><td>✓</td></tr><tr><td>Sensor</td><td>✓</td><td></td><td></td></tr><tr><td>Headphones</td><td></td><td>✓</td><td></td></tr><tr><td>Microphone</td><td>✓</td><td></td><td></td></tr><tr><td>USB flash drive</td><td></td><td></td><td>✓</td></tr><tr><td>Actuator</td><td></td><td>✓</td><td></td></tr></table>				Hardware device	Input (✓)	Output (✓)	Storage (✓)	Solid state drive (SSD)			✓	Sensor	✓			Headphones		✓		Microphone	✓			USB flash drive			✓	Actuator		✓		6
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Question	Answer	Marks
2(b)	– Input – Black – White – Sensors – Binary	5
One mark for each correct term in the correct place		

Question	Answer	Marks
5(a)	Any four from: – Stock control system has a database of stock – Each product has a (unique) barcode – Barcode is scanned, and product looked up in database – Stock levels for product are reduced (by 1) – Stock is checked against minimum level – If stock at/below minimum level an order is placed – When stock is re-ordered flag is reset	4

5(b)	- It has RAM to store the data / programs / by example currently in use - It has ROM to permanently store the boot up instructions - It has HDD to store the stock database / software / OS / by example	3
5(c)	Any four from: - MAR - MDR - PC - ALU - CU - ACC - CIR - Buses - Registers	4

Question	Answer				Marks																												
8	<table><tr><th>Statement</th><th>3D (✓)</th><th>Inkjet (✓)</th><th>Laser (✓)</th></tr><tr><td>Uses a moving print head</td><td>✓</td><td>✓</td><td></td></tr><tr><td>Uses liquid ink</td><td></td><td>✓</td><td></td></tr><tr><td>Produces output using materials such as plastic and resin</td><td>✓</td><td></td><td></td></tr><tr><td>Uses piezoelectric or thermal technology</td><td>(✓)</td><td>✓</td><td></td></tr><tr><td>Uses a rotating drum to transfer the image to the paper</td><td></td><td></td><td>✓</td></tr><tr><td>Uses layer upon layer of material to create the output</td><td>✓</td><td></td><td></td></tr></table>				Statement	3D (✓)	Inkjet (✓)	Laser (✓)	Uses a moving print head	✓	✓		Uses liquid ink		✓		Produces output using materials such as plastic and resin	✓			Uses piezoelectric or thermal technology	(✓)	✓		Uses a rotating drum to transfer the image to the paper			✓	Uses layer upon layer of material to create the output	✓			6
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One mark per each correct row.																																	

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Question	Answer				Marks																												
1	One mark per each correct row: <table><tr><th>Device</th><th>Input (✓)</th><th>Output (✓)</th><th>Storage (✓)</th></tr><tr><td>Keyboard</td><td>✓</td><td></td><td></td></tr><tr><td>Sensor</td><td>✓</td><td></td><td></td></tr><tr><td>3D Cutter</td><td></td><td>✓</td><td></td></tr><tr><td>2D Scanner</td><td>✓</td><td></td><td></td></tr><tr><td>Microphone</td><td>✓</td><td></td><td></td></tr><tr><td>Hard disk drive (HDD)</td><td></td><td></td><td>✓</td></tr></table>				Device	Input (✓)	Output (✓)	Storage (✓)	Keyboard	✓			Sensor	✓			3D Cutter		✓		2D Scanner	✓			Microphone	✓			Hard disk drive (HDD)			✓	6
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Question	Answer				Marks
2(b)	Seven from: - Camera captures code // Laser/light shone on code - Black squares reflect different light to white - Corner squares are used for alignment - Pattern converted to digital data // by example (Digital) data sent to microprocessor - There is a database of valid QR codes - Data compared to stored values/valid QR codes If data matches entry is granted is raised ... If data matches count is incremented ... If data does not match, entry is denied				7

6(b)(i)	– Control (bus)				1															
6(b)(ii)	– Accumulator (ACC)				1															
6(b)(iii)	<table><tr><th>Statement</th><th>True (✓)</th><th>False (✓)</th></tr><tr><td>Data and instructions are stored in the same memory unit</td><td>✓</td><td></td></tr><tr><td>The control unit manages operations within the CPU</td><td>✓</td><td></td></tr><tr><td>Data and instructions can be fetched into the CPU at the same time</td><td></td><td>✓</td></tr><tr><td>The control unit is responsible for decoding an instruction</td><td>✓</td><td></td></tr></table>				Statement	True (✓)	False (✓)	Data and instructions are stored in the same memory unit	✓		The control unit manages operations within the CPU	✓		Data and instructions can be fetched into the CPU at the same time		✓	The control unit is responsible for decoding an instruction	✓		4
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	The control unit is responsible for decoding an instruction	✓																		

Question	Answer	Marks
8(a)	Any three from: – Light emitting diodes (technology) – The display is made up of pixels – ... that are arranged together as a matrix – ... each is formed of three LEDs/filters – Shades of colour are achieved by mixing red, blue and green – The screen can be back-lit/edge-lit NOTE: Use of liquid crystals with LED technology can also be awarded	3

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Question	Answer	Marks
8(b)	Any three from: – Energy efficient // low power consumption – Long lasting // longevity – Focussed beam // less light strays from beam – Brighter/vivid colours – High resolution – No flicker – Display is thinner – Mercury free technology // environmentally friendly – Fewer pixel failure – Increased viewing in sunlight	3
8(c)	– LCD	1

1(c)(i)	– Input	1
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3(b)	Max three from: – Solid state drive – Non-volatile – Secondary storage – Flash memory – Has no mechanical/moving parts – Uses transistors – ... and cells that are laid out in a grid – Uses control gates and floating gates – Can be NAND/NOR (technology) – Use EEPROM technology Max two from: – Stores data by flashing it onto the chips – Data stored by controlling the flow of electrons through/using transistors/chips/gates – The electric current reaches the control gate and flows through to the floating gate to be stored – When data is stored the transistor is converted from 1 to 0	4
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Question	Answer	Marks																												
3(c)	One mark for each correct row: <table><tr><th>Statement</th><th>Blu-ray (✓)</th><th>CD (✓)</th><th>DVD (✓)</th></tr><tr><td>A type of optical storage</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>Has the largest storage capacity</td><td>✓</td><td></td><td></td></tr><tr><td>Can be dual layer</td><td>✓</td><td></td><td>✓</td></tr><tr><td>Read using a red laser</td><td></td><td>✓</td><td>✓</td></tr><tr><td>Has the smallest storage capacity</td><td></td><td>✓</td><td></td></tr><tr><td>Stores data in a spiral track</td><td>✓</td><td>✓</td><td>✓</td></tr></table>	Statement	Blu-ray (✓)	CD (✓)	DVD (✓)	A type of optical storage	✓	✓	✓	Has the largest storage capacity	✓			Can be dual layer	✓		✓	Read using a red laser		✓	✓	Has the smallest storage capacity		✓		Stores data in a spiral track	✓	✓	✓	6
Statement	Blu-ray (✓)	CD (✓)	DVD (✓)																											
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Question	Answer	Marks																		
5(a)	One mark for each correct row: <table border="1"> <thead> <tr> <th>Statement</th><th>True (✓)</th><th>False (✓)</th></tr> </thead> <tbody> <tr> <td>It is a flat panel display</td><td>✓</td><td></td></tr> <tr> <td>It creates images using red, green and blue diodes</td><td>✓</td><td></td></tr> <tr> <td>It is not very energy efficient and gives off heat</td><td></td><td>✓</td></tr> <tr> <td>It is also used in mobile devices such as smartphones and tablets</td><td>✓</td><td></td></tr> <tr> <td>It is a front-lit display</td><td></td><td>✓</td></tr> </tbody> </table>	Statement	True (✓)	False (✓)	It is a flat panel display	✓		It creates images using red, green and blue diodes	✓		It is not very energy efficient and gives off heat		✓	It is also used in mobile devices such as smartphones and tablets	✓		It is a front-lit display		✓	5
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It is a front-lit display		✓																		

5(b)	One mark for each correct term in the correct place: – Control – Unique – Identify – Protocol – Dynamic	5
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0478/13 – NOVEMBER 2020

Question	Answer	Marks																								
1	One mark for each correct row: <table><thead><tr><th>Device</th><th>Input (✓)</th><th>Output (✓)</th><th>Storage (✓)</th></tr></thead><tbody><tr><td>Solid state drive (SSD)</td><td></td><td></td><td>✓</td></tr><tr><td>Headphones</td><td></td><td>✓</td><td></td></tr><tr><td>2D cutter</td><td></td><td>✓</td><td></td></tr><tr><td>LCD projector</td><td></td><td>✓</td><td></td></tr><tr><td>Microphone</td><td>✓</td><td></td><td></td></tr></tbody></table>	Device	Input (✓)	Output (✓)	Storage (✓)	Solid state drive (SSD)			✓	Headphones		✓		2D cutter		✓		LCD projector		✓		Microphone	✓			5
Device	Input (✓)	Output (✓)	Storage (✓)																							
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Question	Answer	Marks
2(a)	– Control unit // CU	1
2(b)	– Arithmetic logic unit // ALU	1
2(c)	– Program counter // memory address register // PC // MAR	1
2(d)	– Memory data register // current instruction register // MDR // CIR	1
2(e)	– Memory data register // MDR	1

Question	Answer	Marks
8(a)	Any three from: – Liquid crystal display – The display is made of pixels – ... arranged in a matrix – Uses a flat panel display – Backlit display – ... with CCFLs/LEDs – Uses light-modulating properties of liquid crystals – Crystals can be turned between opaque and transparent (to allow light to pass) – Colours created using RGB	3
8(b)	Any three from: – Low power consumption – Runs at cool temperature – Do not suffer image burn – Do not suffer flicker issues – Bright image/colours – High resolution image – Cheaper to purchase than e.g. LED screen	3

Question	Answer	Marks
9(a)	Any three from: - Both need a red laser to read/write data - Both are spun to be read - Both use spiral tracks for data - Both are optical storage - Both are off-line storage // both non-volatile - Both use pits and lands to store data	3
9(b)	Any one from: - DVD can be dual layer, but CD can only be single - DVD has higher storage capacity - DVD has a shorter wavelength laser - DVD are spun faster - DVDs have a higher data transfer rate	1

11	Six from: - Suitable sensor used e.g. motion sensor/pressure sensor (Analogue) data is converted to digital (using ADC) - Sensor sends data to microprocessor - Data compared to stored data if value outside range/within range water is sprayed ... signal sent to actuator to spray water ... if value within range/outside range no action taken - Continuous loop	6
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Question	Answer	Marks																		
12	One mark for each correct row: <table border="1"> <thead> <tr> <th>Statement</th><th>True (✓)</th><th>False (✓)</th></tr> </thead> <tbody> <tr> <td>Provides access to the Internet for customers</td><td>✓</td><td></td></tr> <tr> <td>Can determine the maximum bandwidth available for customers</td><td>✓</td><td></td></tr> <tr> <td>Monitors the volume of data downloaded by customers</td><td>✓</td><td></td></tr> <tr> <td>Can provide an IP address for the customer</td><td>✓</td><td></td></tr> <tr> <td>Stores the content for all web pages available on the Internet</td><td></td><td>✓</td></tr> </tbody> </table>	Statement	True (✓)	False (✓)	Provides access to the Internet for customers	✓		Can determine the maximum bandwidth available for customers	✓		Monitors the volume of data downloaded by customers	✓		Can provide an IP address for the customer	✓		Stores the content for all web pages available on the Internet		✓	5
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Question	Answer	Marks										
3	One mark for each device/description <table><tr><th>Name of device</th><th>Description</th></tr><tr><td><u>Inkjet Printer</u></td><td>Uses either thermal bubble or piezoelectric technology</td></tr><tr><td>Actuator</td><td>– Operated by signals to cause a physical movement Controls the movement of a machine // by example</td></tr><tr><td>DLP//Projector</td><td>Uses thousands of tiny mirrors that can move very quickly to create an image</td></tr><tr><td>Mouse</td><td>– Uses rolling ball / optical sensor / laser to detect motion // by example – Movement echoed on screen // moves curser/pointer (on screen) – Has scroll wheel / Buttons to allow data input // by example</td></tr></table>	Name of device	Description	<u>Inkjet Printer</u>	Uses either thermal bubble or piezoelectric technology	Actuator	– Operated by signals to cause a physical movement Controls the movement of a machine // by example	DLP//Projector	Uses thousands of tiny mirrors that can move very quickly to create an image	Mouse	– Uses rolling ball / optical sensor / laser to detect motion // by example – Movement echoed on screen // moves curser/pointer (on screen) – Has scroll wheel / Buttons to allow data input // by example	4
Name of device	Description											
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Question	Answer	Marks
4(a)	Six from Max four from: – Scanned using a barcode reader – Shines (red) laser/light – Light is reflected back // white lines reflect light // black lines reflect less light – Sensors/photoelectric cells detect the light – Different reflections/bars give different binary/digital values // (pattern) converted to binary/digital values – Microprocessor interprets the data – Uses check digit error checking Max three from: – Database stores data/barcodes/products/prices – Barcode/value/key transmitted to database/system // Searches for barcode/value/key in the database/system... – ... price is returned/found	6

Question	Answer	Marks
4(b)(i)	Max three from: – Flash storage – Uses transistors/controls gates/floating gates – Can be NAND/NOR technology // Can use flip-flops – Stores data by flashing it onto the chips/device – Controlling/using the flow of electrons through/using transistors/chips/gates – The electric current reaches the control gate and flows through to the floating gate to be stored – When data is stored the transistor is converted from 1 to 0 / 0 to 1	3
4(b)(ii)	Max one from: – Faster read/write operation – Produces less heat // needs less cooling – Less susceptible to interference/magnets – Less power consumption	1

Question	Answer	Marks
8(a)	– Instructions and data stored in the same/main memory – Instructions fetched and executed in order / one after another / in sequence	2
8(b)(i)	– Holds the address ... – ... of next / current instruction	2
8(b)(ii)	Any two from: – Carries / transfers control signals/instructions // carries/transfers commands ... – ... from CPU/CU to components // from devices to CPU/CU – To synchronise the FE cycle	2

1(b)(i)	Any one from: - Both addresses can be used to identify a computer/device - Both are unique - Both can be represented as hexadecimal - Both addresses do not change if IP address is static	1
1(b)(ii)	Any two from: - An IP address is assigned by the network/router/ISP, A MAC address is assigned by the manufacturer - An IP address can be changed (if dynamic), MAC address cannot be changed - IP address has 4/8 groups of values, MAC address has 6 groups/pairs of values - IP address is 32-bit/128-bit, MAC address is 48-bit - IP address does not contain serial number/manufacture number, MAC address does - IP(v4) address is denary and MAC address is hexadecimal	2

Question	Answer	Marks
2(a)	Any three from: - Keyboard - Mouse - Microphone - Keypad - Touchscreen - Touchpad	3

Question	Answer	Marks																												
2(b)	One mark for each correct row. <table><tr><th>Statement</th><th>HDD (✓)</th><th>SSD (✓)</th><th>USB flash memory drive (✓)</th></tr><tr><td>it has no moving parts</td><td></td><td>✓</td><td>✓</td></tr><tr><td>it is non-volatile</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>it can use NAND gates to store data</td><td></td><td>✓</td><td>✓</td></tr><tr><td>it uses magnetic properties to store data</td><td>✓</td><td></td><td></td></tr><tr><td>it has the smallest physical size</td><td></td><td></td><td>✓</td></tr><tr><td>it has the slowest read/write speeds</td><td>✓</td><td></td><td></td></tr></table>	Statement	HDD (✓)	SSD (✓)	USB flash memory drive (✓)	it has no moving parts		✓	✓	it is non-volatile	✓	✓	✓	it can use NAND gates to store data		✓	✓	it uses magnetic properties to store data	✓			it has the smallest physical size			✓	it has the slowest read/write speeds	✓			6
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Question	Answer	Marks
5	Eight from: – Sensor send data/readings/signal to microprocessor – Data is converted from analogue to digital (using ADC) – Microprocessor compares/checks data to stored values/range of values ... – ... If data is greater than 30 / above the range microprocessor sends signal to open window and to turn heater off – ... If data is below 25 the microprocessor sends signal to turn on heater and to close window – ... If data is between 25 and 30 / within the range no action taken – Actuator is used to operate heater/window – Whole process is continuous	8

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Question	Answer				Marks
2(a)	One mark per each correct row.				6
	Statement	Magnetic (✓)	Solid state (✓)	Optical (✓)	
	no moving parts are used to store data		✓		
	pits and lands are used to store data			✓	
	data is stored on platters	✓			
	flash memory is used to store data		✓		
	parts are rotated to store data	✓		✓	
	data can be stored permanently	✓	✓	✓	
2(b)(i)	Any one from: – Hard disk drive // HDD – Magnetic tape				1

Question	Answer	Marks
2(b)(ii)	Any one from: – CD – DVD – Blu-ray disk	1
2(b)(iii)	One for type of storage, two for matching justification from: – Magnetic // HDD – (Web server) is likely to receive many requests a day – (Web server) will likely need to store a lot of data and magnetic is high capacity – Magnetic is cheaper to buy for storage per unit than solid state – Magnetic is capable of more of read/write requests over time // has more longevity // SSD has more limited number of read/write requests (before it is no longer usable) – No requirement for it to be portable, so moving parts does not matter – Solid-state // SSD – (Web server) is likely to receive many requests a day – (Web server) will likely need to store a lot of data and solid-state is high capacity – Solid-state is more energy efficient – Solid-state runs cooler so will not overheat – Solid state has faster read/write speeds to handle volume of traffic	3
2(c)	Any three from: – Data is flashed onto (silicon) chips – Uses NAND/NOR technology // can use flip-flops – Uses transistors/control gates/floating gates ... – ... to control the flow of electrons – It is a type of EEPROM technology – When data is stored the transistor is converted from 1 to 0 / 0 to 1 – Writes (and reads) sequentially	3
5(a)	– Light sensor – Motion sensor // infra-red sensor	2
5(b)	Eight from: – Sensors send data to microprocessor – Data is converted to digital (using ADC) – Microprocessor compares data to stored value(s) ... – ... if one value or neither values are within range/out of range/match no action is taken – ... If both values are out of range/in range/match microprocessor sends signal to switch light on ... – ... 1-minute timer is started – Actuator used to switch on/off light – When timer reaches 1 minute, microprocessor sends signal to switch light off – Whole process is continuous	8

8	One mark for each correct term in the correct order – Switch – Circuit – Current – Calculated – Character – Binary	6
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1(b)(i)	– Magnetic – Solid state – Optical	3
1(b)(ii)	Any four from: – It has platters – Platters/disk divided into tracks – Platter/disk is spun – Has a read/write arm that moves across storage media – Read/writes data using electromagnets – Uses magnetic fields to control magnetic dots of data – Magnetic field determines binary value	4

Question	Answer	Marks
3(a)	One mark per each correct term, in the correct place. – LED – Photoelectric – Lens – Magnifies – Microswitch – USB	6
3(b)	Any two from: – Keyboard – Microphone – 2D/3D Scanner – Sensor – Touchscreen – Keypad – Webcam – Joystick	2

Question	Answer	Marks
9(a)	One mark per each correct sensor	3

Question	Answer				Marks
10(a)	One mark per each correct row				6
	Statement	ALU (✓)	CU (✓)	RAM (✓)	
	Stores data and instructions before they enter the central processing unit (CPU)			✓	
	Contains a register called the accumulator	✓			
	Manages the transmission of data and instructions to the correct components		✓		
	Contained within the CPU	✓	✓		
	Uses the data bus to send data into or out of the CPU	(✓)		✓	
	Carries out calculations on data	✓			
10(b)	Any two from: – MAR – MDR // MBR – PC – CIR // IR				2

Question	Answer	Marks
2(a)	– Microphone	1
2(b)	– capacitive	1
2(c)	– interrupt	1

4(c)	– Light – Lens – Charge-coupled – Analogue-to-digital – Pixel	5
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Question	Answer	Marks																												
6(a)	One mark per each correct row. <table><tr><th>Statement</th><th>MAR (✓)</th><th>MDR (✓)</th><th>PC (✓)</th></tr><tr><td>it is a register in the CPU</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>it holds the address of the next instruction to be processed</td><td>(✓)</td><td></td><td>✓</td></tr><tr><td>it holds the address of the data that is about to be fetched from memory</td><td>✓</td><td></td><td>(✓)</td></tr><tr><td>it holds the data that has been fetched from memory</td><td></td><td>✓</td><td></td></tr><tr><td>it receives signals from the control unit</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>it uses the address bus to send an address to another component</td><td>✓</td><td></td><td>✓</td></tr></table>	Statement	MAR (✓)	MDR (✓)	PC (✓)	it is a register in the CPU	✓	✓	✓	it holds the address of the next instruction to be processed	(✓)		✓	it holds the address of the data that is about to be fetched from memory	✓		(✓)	it holds the data that has been fetched from memory		✓		it receives signals from the control unit	✓	✓	✓	it uses the address bus to send an address to another component	✓		✓	6
Statement	MAR (✓)	MDR (✓)	PC (✓)																											
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it uses the address bus to send an address to another component	✓		✓																											
6(b)	– Arithmetic Logic Unit // ALU	1																												

Question	Answer	Marks
7(a)	One mark per correct storage, two marks for justification. – Secondary – It is non-volatile storage – It is not directly accessed by the CPU	3
7(b)	Any four from: – Uses flash memory – Data is flashed onto (silicon) chips – Uses NAND/NOR technology // Can use flip-flops – Uses transistors/control gates/floating gates ... – ... to control the flow of electrons – It is a type of EEPROM technology – When data is stored the transistor is converted from 1 to 0 / 0 to 1 – Writes (and reads) sequentially	4

Question	Answer	Marks								
9(a)	One mark per each correct sensor. <table><tr><th>Task</th><th>Sensor</th></tr><tr><td>checking the water is 30 °C</td><td>Temperature</td></tr><tr><td>checking the water acidity level after detergent is added</td><td>pH</td></tr><tr><td>checking the weight of the clothes to make sure that the machine is not overloaded</td><td>Pressure</td></tr></table>	Task	Sensor	checking the water is 30 °C	Temperature	checking the water acidity level after detergent is added	pH	checking the weight of the clothes to make sure that the machine is not overloaded	Pressure	3
Task	Sensor									
checking the water is 30 °C	Temperature									
checking the water acidity level after detergent is added	pH									
checking the weight of the clothes to make sure that the machine is not overloaded	Pressure									
9(b)	Six from: – Sensor sends data to microprocessor – Data is converted from analogue to digital (using ADC) – Data is compared to stored value (of 30) If data is below 30 then a microprocessor sends signal is sent to a heater to heat the water up/add hot water – if data is above 30 then a microprocessor sends signal is sent to turn the heater off to allow the water to cool down/add cold water – Actuator used to turn headset on/off // Actuator used to add water – If data is 30 then no action is taken – It is a continuous process	6								

NOVEMBER 2021 (2)

Question	Answer	Marks
2(a)	Any one from: – Printer – Speaker – Light/LED – Actuator	1
2(b)	Any one from: – Touchscreen – Trackpad / touchpad – Microphone – QR code reader – Barcode reader – Magnetic strip reader – RFID reader	1

Question	Answer	Marks
4	One mark per each correct term in the correct order. – Capacitive – Conductive // Capacitive – Change – Coordinates – Resistive – Circuit – Manufacture	7

Question	Answer				Marks
7	One mark per each correct row.				5
	Statement	3D scanner (✓)	Barcode reader (✓)	QR code reader (✓)	
	uses position and alignment markers for orientation when scanning			✓	
	scans the shape and appearance of an object	✓			
	uses reflected light from a laser to convert a black-and-white pattern into binary		✓	(✓)	
	can often be built into an Electronic Point Of Sale (EPOS) terminal, for example, a supermarket checkout		✓	(✓)	
	it is an example of an input device	✓	✓	✓	

Question	Answer	Marks
8	Seven from: – Timer is started – Pressure sensor (within each mat) – Sensor sends data to microprocessor – Analogue data is converted to digital (using ADC) – Microprocessor compares data to stored value(s) – If data matches / in/out range microprocessor stops timer – If data matches / in/out range microprocessor checks if data has come from correct colour mat sensor – If data matches / in/out range microprocessor checks to see if timer is stopped at less than 1 second – If data matches / in/out range microprocessor increments counter if timer is less than 1 second and colour/mat is correct – If correct colour/mat is hit, timer is reset and the whole process is repeated – If data has not come from the correct colour mat sensor the game ends	7

2(d)(i)	– Pressure sensor – Motion sensor	2								
2(d)(ii)	One mark for the correct tick <table><tr><th>Device</th><th>Tick (✓)</th></tr><tr><td>input</td><td>✓</td></tr><tr><td>storage</td><td></td></tr><tr><td>output</td><td></td></tr></table>	Device	Tick (✓)	input	✓	storage		output		1
Device	Tick (✓)									
input	✓									
storage										
output										

Question	Answer	Marks														
5	One mark per correct term or description. <table><tr><th>Component name</th><th>Description</th></tr><tr><td>Memory Address Register (MAR)</td><td>(A register that) holds the address of the data/instruction that needs to be fetched/processed // holds the address of where the data needs to be stored.</td></tr><tr><td>Program Counter (PC)</td><td>(A register that) holds the address of the next / current instruction to be processed.</td></tr><tr><td>accumulator // ACC</td><td>This is a register that is built into the arithmetic logic unit. It temporarily holds the result of a calculation.</td></tr><tr><td>memory data register // MDR</td><td>This is a register that holds data or an instruction that has been fetched from memory.</td></tr><tr><td>Control Unit (CU)</td><td>Sends control signals to control the flow of data through the CPU // manages the execution of instructions in the CPU</td></tr><tr><td>address bus</td><td>This carries addresses around the CPU.</td></tr></table>	Component name	Description	Memory Address Register (MAR)	(A register that) holds the address of the data/instruction that needs to be fetched/processed // holds the address of where the data needs to be stored.	Program Counter (PC)	(A register that) holds the address of the next / current instruction to be processed.	accumulator // ACC	This is a register that is built into the arithmetic logic unit. It temporarily holds the result of a calculation.	memory data register // MDR	This is a register that holds data or an instruction that has been fetched from memory.	Control Unit (CU)	Sends control signals to control the flow of data through the CPU // manages the execution of instructions in the CPU	address bus	This carries addresses around the CPU.	6
Component name	Description															
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address bus	This carries addresses around the CPU.															

MARCH 2022

Question	Answer	Marks
4(a)(i)	1 mark for each completed statement An optical mouse shines a red light from a Light-Emitting Diode//LED underneath the mouse. The light reflects back from a surface through a lens in the mouse and is converted to a value. This value is transmitted to the computer. The computer then determines the direction and speed of the movement. When the user presses a key on a keyboard, the key pushes the switch on the circuit board. This completes a circuit. Signals are sent to the computer that uses the data to calculate which key was pressed.	6
4(a)(ii)	1 mark each e.g. • touchscreen • touchpad • scanner • microphone	2
4(b)	Any three from: • More visible pixels // higher resolution • Higher colour contrast (in ambient lighting) // more vivid colours • Colours are (often) more accurate • Image (usually) appears brighter (with same wattage) • Will be stationary so does not need the portability of DLP • Does not need the compactness of DLP • Cost of purchase (usually) less • Run quieter • Any surface can be used as a display • Uses less power • Produces less heat • Does not give the rainbow effect DLP often gives • Longer lasting lamps	3

Question	Answer	Marks
4(c)	1 mark for naming appropriate device • hard disk / HDD / magnetic • solid state drive / SSD / USB memory stick / flash drive 1 mark each for matching justification to max 3. e.g. HDD • Large capacity storage • ...for videos/films that have large file sizes / require large storage space • Longevity // Can be read from/written to large number of times • ... the copies will be accessible for a long time // device will not break if accessed regularly • ...may require large number of read/write to access fields • Relatively cheap per GB // cheaper than SSD per GB ... • ... there is a large capacity required therefore overall may cost less than other devices • Device will not be moved regularly so no need for portability/durability SSD • Large capacity storage • ...for videos/films that have large file sizes / require large storage space • No moving parts • ...can be carried/moved to other locations with limited risk of damage • Fast access speed • ... videos are large files that will be stored/accessed in less time • Cost per GB is not significant (in comparison) • ...there is a large capacity required therefore overall cost will not be significantly more than others • Used as a copy of files to may not be accessed regularly • ... the limited number of read/write times/longevity is inconsequential • Uses less power // runs cooler • No latency // does not take time to start-up	4

Question	Answer	Marks
6	Any seven from: • (Motion) sensor sends signals to microprocessor • analogue signal is converted to digital • microprocessor compares signal to stored value ... • ...if it does not meet / meets the stored value (and if camera is not recording) the microprocessor sends signal (to camera) to start recording • ... if it does not meet / meets the stored value the microprocessor starts/resets the timer • When the timer reaches 2 minutes the microprocessor sends signal (to camera) to stop recording • Whole process is repeated continually/until turned off	7

Question	Answer	Marks
2(a)(i)	Random access memory // RAM	1
2(a)(ii)	One mark for each correct stage Second stage - Decode Third stage - Execute	2
2(a)(iii)	Any two from: - Memory address register // MAR - Memory data register //MDR - Program counter // PC - Control unit // CU - Address bus - Data bus - Control bus	2

Question	Answer	Marks
3(a)	Any one from e.g.: - SD card - USB flash memory drive - Random access memory // RAM	1
3(b)	Any three from: - Laser is shone at the disk - An (arm/head) moves the laser across the surface of the disk - The laser burns pits onto the surface of the disk - The laser is used to read the pits and lands on the surface of the disk - The reflected light from the laser shining on the disk is captured (by a sensor)	3
3(c)(i)	Any three from: - It is small in size // compact // lightweight // portable - It has low power consumption - It runs quietly - It runs at a cool temperature - It is robust when dropped // durable - Fast access/read/write speed - High capacity - Not affected by magnets	3
3(c)(ii)	- Operating system - Application software // by example - Utility software // by example	2

Question	Answer	Marks
10	Any three from: - It is a unique address - It is assigned by the manufacturer - It can be used to identify a device - It contains the manufacturer ID/code/number - It contains the serial code/number - It is written in hexadecimal - It has 6 bytes/48 bits/6 pairs of digits - Does not (usually) change // static	3

Question	Answer	Marks
11(a)	One mark for each of the correct terms, in the correct place • buffer • nozzles • liquid • thermal bubble • interrupt	5
11(b)	Any three from e.g.: • Monitor // screen • Speaker • Headphones • Light // LED • (2D/3D) cutter • DLP // LCD projector • Actuator	3
11(c)	Any two from e.g.: • Keyboard • Mouse // trackball • Microphone • (2D/3D) scanner • Barcode reader • QR code reader • Digital camera // webcam • Interactive whiteboard • Touchscreen • Sensor	3

JUNE 2022(2)

Question	Answer	Marks																																
7(a)	Three from: • CD • DVD • Blu-ray	3																																
7(b)	One mark for each correct row <table><tr><th></th><th colspan="3">Type of storage</th></tr><tr><th>Statement</th><th>Magnetic (✓)</th><th>Optical (✓)</th><th>Solid state (✓)</th></tr><tr><td>this storage has no moving parts</td><td></td><td></td><td>✓</td></tr><tr><td>this storage uses a laser to read and write data</td><td></td><td>✓</td><td></td></tr><tr><td>this storage uses a read/write head</td><td>✓</td><td>✓</td><td></td></tr><tr><td>this storage burns pits onto a reflective surface</td><td></td><td>✓</td><td></td></tr><tr><td>this storage uses NAND and NOR technology</td><td></td><td></td><td>✓</td></tr><tr><td>this storage stores data in tracks and sectors</td><td>✓</td><td>(✓)</td><td></td></tr></table>		Type of storage			Statement	Magnetic (✓)	Optical (✓)	Solid state (✓)	this storage has no moving parts			✓	this storage uses a laser to read and write data		✓		this storage uses a read/write head	✓	✓		this storage burns pits onto a reflective surface		✓		this storage uses NAND and NOR technology			✓	this storage stores data in tracks and sectors	✓	(✓)		6
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Question	Answer	Marks																								
1(a)	One mark for each correct row	5																								
	<table><tr><th>Device</th><th>Input (✓)</th><th>Output (✓)</th><th>Storage (✓)</th></tr><tr><td>keyboard</td><td>✓</td><td></td><td></td></tr><tr><td>microphone</td><td>✓</td><td></td><td></td></tr><tr><td>headphones</td><td></td><td>✓</td><td></td></tr><tr><td>hard disk drive (HDD)</td><td></td><td></td><td>✓</td></tr><tr><td>actuator</td><td></td><td>✓</td><td></td></tr></table>		Device	Input (✓)	Output (✓)	Storage (✓)	keyboard	✓			microphone	✓			headphones		✓		hard disk drive (HDD)			✓	actuator		✓	
	Device		Input (✓)	Output (✓)	Storage (✓)																					
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actuator		✓																								
1(b)	- Optical - Solid state	2																								

Question	Answer	Marks
2(b)	Any two from: - It has a bright screen/colours - It does not consume much power - It runs at a cool temperature - Longevity - Can operate in cold conditions - Can be small/compact/thin	2

Question	Answer	Marks
5(a)	Any two from: - It processes data - It processes/executes instructions - It carries out calculations - It carries out logical operations Any three from: - Fetch - Decode - Execute	5

Question	Answer	Marks
5(b)(i)	Any four from: - Memory data register / MDR - Control unit / CU - Arithmetic logic unit / ALU - Program counter / PC - Current instruction register / Instruction register / CIR / IR - Accumulator / ACC - Address bus - Data bus - Control bus	4
5(b)(ii)	Any two from: - To temporarily store ... the address of the next data/instructions required ... the address of the location in memory where data is to be stored - Stores the address that is to be collected by the address bus	2

Question	Answer	Marks
5(c)(i)	Any four from: • It is RAM ... • ... which is volatile storage • It is ROM ... • ... which is non-volatile storage • It is directly accessible by the CPU Any three from: • BIOS • Start-up instructions • Programs that are currently in use • Data that is currently in use • Parts of OS currently in use	7
5(c)(ii)	Any two from: • Data can be permanently stored • ... meaning that (application) software can be loaded/retrieved • ... meaning that operating system can be loaded/retrieved • ... meaning that user data/files can be accessed/retrieved	2

6(c)	Any four from: • A light is shone onto the surface of the document • The light is moved across/down/under the document • The reflected light is captured (using mirrors and lenses) • The reflections are converted to binary	4
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3(b)	Any three from: - Currently running data - Currently running (application) software - Currently running instructions - Currently running parts of OS - Currently running utility software	3
3(c)	Any two from: - For non-volatile/permanent/long-term storage of files/data - To store data that is not currently required by the CPU - To store data to transfer it to another computer	2

4(c)	One mark for each term: Solid-state storage is non-volatile. This means that the data is not lost when the power is turned off. Solid-state storage is made of transistors that are laid out in a grid. Gates are used to control the flow of the electrons through the transistors. This changes the data in the transistors from 1 to 0, or from 0 to 1.	4
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Question	Answer	Marks
6(c)	Seven from: - Uses an infra-red/proximity sensor - Sensor continuously sends the digitised value/reading/distance to the microprocessor - Microprocessor compares the data/signal to the stored value of 10(cm) - If the data/signal is greater than the stored value/10a signal is sent to make the robot move forward - If the data/signal is less than or equal to the stored value/10a signal is sent to make robot turn - An actuator is used to make the robot turn/move forward - The whole process repeats continuously until turned off/stopped	7

Question	Answer	Marks
3(a)	One mark for each correctly circled storage device: • Compact disk (CD) • Solid-state drive (SSD) • Hard disk drive (HDD)	3
3(b)	• C	1

Question	Answer	Marks
5(a)	Any one from: • Level • Pressure • Moisture	1
5(b)	Any Six from: • Sensor continually sends digitised data to microprocessor • Microprocessor compares data to stored value(s) • If value is outside range / matches microprocessor sends signal to release water to refill water bowl • ... bowl filled by set amount // bowl filled for certain time • Actuator used to release water • Whole process repeats until turned off/stopped	6

Question	Answer	Marks
6(a)(i)	• Network interface card/controller // NIC // WNIC	1
6(a)(ii)	• Media access control/MAC address // MAC	1
6(b)(i)	• Router	1
6(b)(ii)	Three from: • It can be used to uniquely identify a device (on a network) • It can change ... • ... each time the device is connected to the network	3

JUNE 2023 (2)

Question	Answer	Marks
1	One mark for each correct device: • Actuator • Printer • Speaker	3

Question	Answer	Marks
4(b)(i)	• Router	1
4(b)(ii)	Any two from: • A collection of servers • ... that store data in a remote location // that allows data to be accessed remotely • ... that are (normally) accessed using an internet connection	2
4(b)(iii)	Any one from: e.g. • May be less secure // by example • May lose access to them if internet connection lost/not available • Reliant on a third party maintaining the hardware // by example • Could incur an extra/ongoing fee/cost	1

Question	Answer	Marks
8(a)	• C	1
8(b)	Four marks from: Any FOUR from: • It is denary based • ... with numbers between 0 and 255 • It is 32 bits • 4 sets/groups of numbers • ... separated by dots Any TWO from: • It is a unique address • It can be static or dynamic • It can be public or private • It contains the network prefix • ... and the host number	4

Question	Answer	Marks
2(a)	Any one from: - To perform a fetch-decode-execute cycle - To process / execute an instruction	1
2(b)	Two from: - It may increase the performance ... because more instructions can be processed simultaneously	2
2(c)(i)	Two from: - To store / holds data / address / instruction ... temporarily	2

Question	Answer	Marks
2(c)(ii)	One mark for correct name of bus. Two marks for matching description. Address bus Transmit / carries addresses between components in the CPU Data bus Transmit / carries data between components in the CPU Control bus Transmits control signals from the control unit to other components in the CPU	3
2(d)	Any two from: e.g. - Keyboard // Keypad - Mouse - Touchscreen - Digital camera - QR code scanner - Barcode scanner 2D scanner - Microphone	2
2(e)	- Any one from: - Speakers - Headphones	1
2(f)	- random access memory (RAM) - read only memory (ROM)	2

Question	Answer	Marks
2(g)	Any three from: - Receives data from the self-checkout system - Compares the book data received to stored book data ... that is a database of stock - If the book is found it decrements the book stock by 1 - If the book is not found an error message is displayed	3

Question	Answer	Marks												
3	One mark for each correct missing term or definition: <table><tr><th>Term</th><th>Definition</th></tr><tr><td>router</td><td>a device that forwards packets to their correct destinations in a network</td></tr><tr><td>IP address</td><td>this address is assigned by the network and used to identify a device on a network</td></tr><tr><td>network interface card (NIC)</td><td>this is a component in a device that enables it to connect to a network</td></tr><tr><td>MAC address</td><td>this address is assigned by the manufacturer and is used to uniquely identify the device</td></tr><tr><td>firewall // proxy-server</td><td>this can be hardware or software based and filters traffic coming into and out of a network</td></tr></table>	Term	Definition	router	a device that forwards packets to their correct destinations in a network	IP address	this address is assigned by the network and used to identify a device on a network	network interface card (NIC)	this is a component in a device that enables it to connect to a network	MAC address	this address is assigned by the manufacturer and is used to uniquely identify the device	firewall // proxy-server	this can be hardware or software based and filters traffic coming into and out of a network	5
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firewall // proxy-server	this can be hardware or software based and filters traffic coming into and out of a network													

Question	Answer	Marks
9(a)	Any one from: • They can both be used to identify a device (on a network) • They can both be static / dynamic • They are both unique (to a device on a network) • They can both be assigned by a router • They can both be public/private	1
9(b)	Four from: • IPv4 is usually written as denary • ... IPv6 usually written as hexadecimal • IPv4 is separated using dots • ... Pv6 is separated using colons • IPv4 is 32-bit • ... IPv6 is 128-bit • IPv4 is 4 groups of digits • ... IPv6 is 8 groups of digits • IPv4 digits are between 0 and 255 • ... IPv6 digits are between 0000 and FFFF • IPv4 all 0s are displayed • ... IPv6 can use double colons to replace repeated groups of 0000 • IPv4 has fewer available unique addresses • ... IPv6 has more available unique addresses	4
9(c)(i)	• Domain name server // DNS	1
9(c)(ii)	• Web browser	1

Question	Answer	Marks
10	Any four from: • The secondary storage / hard drive can be partitioned to create the virtual memory • ... and page B sent to the virtual memory ... • ... which makes space for page D in RAM • ... Once page A / C / D / another page is not required / has been processed • ... page B can be sent from the virtual memory back to RAM when it is required	4

Question	Answer	Marks
1(a)	– A	1
1(b)	– C	1
1(c)	– Control unit	1
1(d)	Any two from: – (The CPU completes) 2.4 billion – ... cycles/clock pulses per second	2
1(e)(i)	Any two from: – Stores data ... – ... that has been fetched/to be written to memory	2
1(e)(ii)	Any three from: – Memory address register // MAR – Program counter // PC – Current instruction register // CIR – Accumulator // ACC	3

2(e)	Any four from: – Compares the ticket number received to stored data – ... that is a database/file of ticket numbers – ... checks the ticket number is listed as paid – If the data matches/cost is paid it sends a signal to raise the barrier – If the data does not match/cost is not paid, the barrier remains down	4
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Question	Answer	Marks
8(a)	Any two from: – Data is stored on platters – (Platters) are divided into tracks and sectors – Has components that are spun – Data is read/written using a read/write arm – Data is read/written using electromagnets – Magnetic field determines the binary value – It is non-volatile	2
8(b)	Any three from: – Flashes data onto chips – Uses transistors – Uses NAND/NOR technology – Uses control gates – Uses flow gates – Controls the flow of electrons – It can be volatile or non-volatile	3
8(c)	One mark for an example in each storage type e.g.: Magnetic – HDD – Magnetic tape – Floppy disk – Magnetic strip Solid state – SSD – USB drive – SD card – RAM Optical – CD – DVD – Blu-ray	3

NOVEMBER 2023 (2)

Question	Answer	Marks
3(a)(i)	– The maximum number of FDE cycles/instructions a CPU can perform/process/execute in a second	1
3(a)(ii)	– Increases/improves the performance // Tasks can be performed quicker/faster – ... because more FDE cycles/instructions can be processed in a second	2
3(b)	– Stores addresses ... – ... of next instruction/data to be fetched // where data is to be written to	2
3(c)	– Instruction set	1

Question	Answer	Marks
4(a)	Any two from: – Performs a single/limited/dedicated function/task – It has a microprocessor – It has dedicated hardware – Uses firmware – It is normally built into a larger device/system – User normally cannot reprogram – It does not require much power – It is cheap to manufacture – Works automatically // works without human intervention – It is small (in size) – It is a real-time system	2
4(b)	One mark for each correct system: – security light system – freezer – vending machine	3

Question	Answer	Marks
7(a)	Any two from e.g. – Barcode scanner – QR code scanner – Digital camera	2
7(b)	Any six from: – Proximity/infrared/pressure sensor used – Sensor continually sends digitised data to microprocessor // When driver pushes button, sensor sends digitised data to the microprocessor – Microprocessor compares data to stored value(s) – If in range/out of range/matches, microprocessor sends signal to close the door – Actuator used to close door – If not in range/out of range/does not match door will not close // – If not in range/out of range/does not match actuator not activated/signal not sent as passenger in door // – If not in range/out of range/does not match a timer is set to check again // – If not in range/out of range/does not match a signal is sent to alert the driver/output a message – This process repeats until the door can close	6

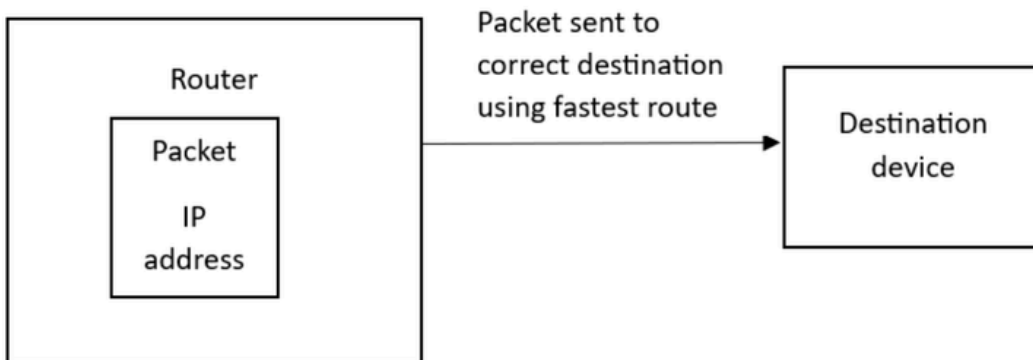
Question	Answer	Marks
13(a)(i)	– C	1
13(a)(ii)	Any one from: – Directly accessed by the CPU – Has both volatile and non-volatile storage	1
13(b)	One mark for each correct term in the correct order: – Random access memory (RAM) – Hard disk drive (HDD) – Pages – Random access memory (RAM) – Virtual memory // Hard disk drive (HDD)	5

NOVEMBER 2023 (3)

Question	Answer	Marks
1(a)	Any two from e.g.: – Touchscreen – Microphone – Keyboard – Keypad – Digital camera – Sensor // by example – Biometric device – Button	2
1(b)	Any one from e.g.: – Screen – Speaker – LED/Light – Actuator/Motor	1

Question	Answer	Marks
3(a)	– Accumulator (ACC) – Control unit (CU) – Program counter (PC)	3
3(b)	Any two from: – It is a type of storage – ...that stores frequently used data/instructions – To speed up access – ... as it is faster to access than RAM – It has different levels e.g. L1 – L3	2
3(c)	– Clock	1
3(d)	– Arithmetic logic unit // ALU	1

Question	Answer	Marks
6(a)	– B	1
6(b)	Four from e.g.: – The data can be accessed from any location – ... meaning that employees can work from anywhere with a connection – The hardware is owned/maintained by a third party – ... meaning that the company are not responsible for maintaining // meaning the company aren't responsible for its security – Can increase the storage needed easily – ... without needing to buy new hardware – Do not need to house the hardware needed – ... costs can be saved on the space saved for this – Cloud system will back up the data – ... meaning the company does not need to do this	4
6(c)	Two from e.g.: – <u>Internet</u> connection is needed/needs to be stable ... – ... and if this is not available/unstable the data cannot be accessed – Employees could be pressured to work outside of hours ... – ... as they can access the data from any location	2

8	The diagram demonstrates (one mark for each part): – The router examining the packet ... – ... looks for the packet header – ... looking for the IP address of destination – The packet being sent toward its correct destination – ... by the fastest route // decides which route it takes – Router is shown connecting devices/networks – Router is shown assigning an IP address to a device e.g.  <pre> graph LR subgraph Router_Box [Router] subgraph Packet_Box [Packet] IP[IP address] end end Packet_Box -- "Packet sent to correct destination using fastest route" --> Destination[Destination device] </pre> Routers examines packet to look for header that has the IP address of destination	4
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Question	Answer	Marks												
9(a)	One mark for each correct type: <table><tr><th>Type of secondary storage</th><th>Statement</th></tr><tr><td>optical</td><td>data is stored using pits and lands</td></tr><tr><td>solid-state</td><td>data is stored using control gates and floating gates</td></tr><tr><td>magnetic</td><td>data is stored using electromagnets</td></tr><tr><td>optical</td><td>data is stored using a laser</td></tr><tr><td>magnetic</td><td>data is stored on a platter that is divided into tracks and sectors</td></tr></table>	Type of secondary storage	Statement	optical	data is stored using pits and lands	solid-state	data is stored using control gates and floating gates	magnetic	data is stored using electromagnets	optical	data is stored using a laser	magnetic	data is stored on a platter that is divided into tracks and sectors	5
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optical	data is stored using a laser													
magnetic	data is stored on a platter that is divided into tracks and sectors													
9(b)	Four from: – Primary storage is directly accessible by the CPU – ... whereas secondary storage is not directly accessible by the CPU – Primary storage stores the data that is currently in use/for booting the system – ... whereas secondary storage stores user's files/data/operating system/application software – Primary storage normally has a small capacity – ... whereas secondary storage normally has a larger capacity – Some parts of primary storage are volatile – ... whereas secondary storage is non volatile – Some parts of primary storage the data cannot be changed – ... whereas data in secondary storage can be changed – Primary storage has faster access speeds to data – ... whereas secondary storage has slower access speeds to data	4												